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Question 1

Backpropagation Algorithm in a Multilayer Perceptron for Binary Classification

Introduction

A **Multilayer Perceptron (MLP)** is a type of artificial neural network used for solving classification problems. It consists of an input layer, one or more hidden layers, and an output layer. During training, the **backpropagation algorithm** updates the weights of the network by minimizing the prediction error. This helps the neural network improve its accuracy over multiple iterations.

Step 1: Forward Pass

In the forward pass, the input data is passed through the input layer to the hidden layer and then to the output layer. Each neuron calculates a weighted sum of its inputs and applies an activation function to produce the output.

Example:

- Age = 45
- Blood Pressure = High
- Glucose Level = High
- Lifestyle Risk = High

Expected Output = **Yes**

Predicted Output = **0.80 (Yes)**

Step 2: Error Calculation

After obtaining the predicted output, the error is calculated by comparing it with the actual output.

Formula:

$$\text{Error} = \text{Actual Output} - \text{Predicted Output}$$

Example:

$$\text{Actual Output} = 1$$

$$\text{Predicted Output} = 0.80$$

$$\text{Error} = 1 - 0.80 = 0.20$$

The objective is to reduce this error during training.

Step 3: Backward Propagation

The calculated error is propagated backward through the network. The weights are updated using the gradient descent algorithm.

Weight Update Formula:

$$\text{New Weight} = \text{Old Weight} + (\text{Learning Rate} \times \text{Error} \times \text{Input})$$

Example:

$$\text{Old Weight} = 0.50$$

$$\text{Learning Rate} = 0.10$$

$$\text{Error} = 0.20$$

$$\text{Input} = 1$$

$$\text{New Weight}$$

$$= 0.50 + (0.10 \times 0.20 \times 1)$$

$$= 0.52$$

The network repeats the forward pass and backward pass until the error becomes very small.

Role of Learning Rate

The learning rate controls how much the weights are updated during each iteration.

- A **small learning rate** gives slow but stable learning.
- A **large learning rate** makes learning faster but may skip the optimal solution.
- Choosing an appropriate learning rate improves convergence.

Role of Activation Functions

Activation functions introduce non-linearity into the neural network, enabling it to solve complex classification problems.

Common activation functions include:

- Sigmoid
- ReLU (Rectified Linear Unit)
- Tanh

These functions improve the learning capability of the neural network.

Conclusion

Backpropagation is an important supervised learning algorithm used in neural networks. It updates the weights by minimizing the prediction error, resulting in improved classification accuracy and better model performance.

Question 2

Genetic Algorithm for Optimizing Candidate Solutions

Introduction

A **Genetic Algorithm (GA)** is an optimization technique inspired by natural evolution. It improves candidate solutions by repeatedly applying the operations of **selection**, **crossover**, and **mutation** over several generations until the best solution is found.

Dataset

Candidate Speed Accuracy Fitness

A	70	80	150
B	85	75	160
C	60	65	125
D	90	85	175

$$Fitness = Speed + Accuracy$$

Higher fitness indicates a better solution.

Generation 1

Initial Population:

A, B, C and D

Fitness Values:

- A = 150
- B = 160
- C = 125
- D = 175

Selection

Selection chooses the individuals with the highest fitness values to become parents for the next generation.

Selected Parents:

- D (Fitness = 175)
- B (Fitness = 160)

These individuals have the highest probability of producing better offspring.

Crossover

Crossover combines the characteristics of two parents to generate new offspring.

Parent 1:

Speed = 90

Accuracy = 85

Parent 2:

Speed = 85

Accuracy = 75

Offspring:

Speed = **90**

Accuracy = **75**

Fitness = **165**

Mutation

Mutation randomly changes one or more values of the offspring to introduce diversity into the population.

Mutated Offspring:

Speed = **92**

Accuracy = **75**

Fitness = **167**

Mutation prevents the algorithm from getting trapped in local optimum solutions.

Generation 2

Candidate	Fitness
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D	175
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Mutated Offspring 167

B	160
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The best candidate continues to survive because it has the highest fitness.

Final Optimized Solution

Speed	Accuracy	Fitness
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90	85	175
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This is the optimal solution after evolution.

Functions of Genetic Algorithm

Selection

- Chooses the best individuals based on fitness.
- Increases the chance of producing better offspring.

Crossover

- Combines features from two parents.
- Produces new candidate solutions.

Mutation

- Introduces random changes.
- Increases diversity.
- Helps explore the search space.

Conclusion

The Genetic Algorithm continuously improves the quality of candidate solutions over multiple generations. By applying selection, crossover, and mutation, it successfully identifies the optimal solution with the highest fitness value.

Question 3

Neural Network Classification for Disease Prediction

Introduction

Neural networks are widely used for medical diagnosis and disease prediction. In this dataset, the input features are **Age, Blood Pressure, Glucose Level, and Lifestyle Risk**, while the output indicates whether the patient has a disease (**Yes** or **No**). The network is trained using the **backpropagation algorithm**.

Training Process

Step 1: Data Preprocessing

Before training, the data is prepared by:

- Converting Yes/No values into numerical values.
- Encoding High, Medium, and Low into numbers.
- Normalizing Age values.
- Removing missing or inconsistent data.

This improves the quality of the dataset.

Step 2: Forward Pass

The input features are supplied to the neural network.

The network processes the inputs through hidden layers and predicts whether the patient has the disease.

Step 3: Error Calculation

The predicted output is compared with the actual output.

If the prediction is incorrect, the error is calculated.

Step 4: Backpropagation

The error is propagated backward through the network.

The weights are updated repeatedly until the prediction error becomes minimum.

Confusion Matrix Calculation

From the given dataset:

Example Actual Predicted

1	Yes	Yes
2	Yes	No
3	No	No
4	Yes	Yes
5	No	Yes

True Positive (TP)

Actual = Yes

Predicted = Yes

Examples = 1 and 4

TP = 2

True Negative (TN)

Actual = No

Predicted = No

Example = 3

TN = 1

False Positive (FP)

Actual = No

Predicted = Yes

Example = 5

FP = 1

False Negative (FN)

Actual = Yes

Predicted = No

Example = 2

FN = 1

Performance Evaluation

Accuracy

Formula:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$= (2 + 1) / (2 + 1 + 1 + 1)$$

$$= 3 / 5$$

$$= \mathbf{0.60}$$

Accuracy = 60%

Precision

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$= 2 / (2 + 1)$$

$$= 66.7\%$$

Recall

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$= 2 / (2 + 1)$$

$$= 66.7\%$$

F1-Score

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

$$= 66.7\%$$

Importance of Preprocessing

- Removes missing values.
- Converts categorical values into numerical form.
- Normalizes input features.
- Improves training speed and model accuracy.

Importance of Feature Selection

- Removes irrelevant features.
- Reduces overfitting.
- Improves prediction accuracy.
- Increases model reliability.

Conclusion

The neural network successfully classifies disease prediction using backpropagation. From the confusion matrix, the model achieved **60% accuracy**, **66.7% precision**, **66.7% recall**, and **66.7% F1-score**. Proper preprocessing and feature selection further improve the model's accuracy and reliability.